

Chapter 10

Cholesky Factorization

The matrix of that equation system is negative definite—which is a positive definite system that has been multiplied through by -1 . For all practical geometries the common finite difference Laplacian operator gives rise to these, the best of all possible matrices. Just about any standard solution method will succeed, and many theorems are available for your pleasure.
—FORMAN S. ACTON, *Numerical Methods That Work* (1970)

Many years ago we made out of half a dozen transformers a simple and rather inaccurate machine for solving simultaneous equations—the solutions being represented as flux in the cores of the transformers. During the course of our experiments we set the machine to solve the equations—
$$X + Y + Z = 1$$
$$X + Y + Z = 2$$
$$X + Y + Z = 3$$
The machine reacted sharply—it blew the main fuse and put all the lights out.
—B. V. BOWDEN, *The Organization of a Typical Machine* (1953)

There does seem to be some misunderstanding about the purpose of an a priori backward error analysis. All too often, too much attention is paid to the precise error bound that has been established. The main purpose of such an analysis is either to establish the essential numerical stability of an algorithm or to show why it is unstable and in doing so to expose what sort of change is necessary to make it stable. The precise error bound is not of great importance.
—J. H. WILKINSON, *Numerics/ Linear Algebra on Digits/ Computers* (1974)

10.1. Symmetric Positive Definite

Symmetric positive definiteness is one of the highest accolades to which a matrix can aspire. Symmetry confers major advantages and simplifications in the eigenproblem and, as we will see in this chapter, positive definiteness permits economy and numerical stability in the solution of linear systems.

A symmetrical matrix $A \in \mathbb{R}^{n \times n}$ is *positive definite* if $x^T A x > 0$ for all nonzero $x \in \mathbb{R}^n$. Well-known equivalent conditions to $A = A^T$ being positive definite are

- $\det(A_k) > 0$, $k = 1:n$, where $A_k = A(1:k, 1:k)$ is the leading principal submatrix of order k .
- $\lambda_k(A) > 0$, $k = 1:n$, where λ_k denotes the k th largest eigenvalue.

The first of these conditions implies that A has an LU factorization, $A = LU$ (see Theorem 9.1). Another characterization of positive definiteness is that the pivots in LU factorization are positive, since $u_{kk} = \det(A_k)/\det(A_{k-1})$. By factoring out the diagonal of U and taking its square root, the LU factorization can be converted into a *Cholesky factorization*: $A = R^T R$, where R is upper triangular with positive diagonal elements. This factorization is so important that it merits a direct proof.

Theorem 10.1. *If $A \in \mathbb{R}^{n \times n}$ is symmetric positive definite then there is a unique upper triangular $R \in \mathbb{R}^{n \times n}$ with positive diagonal elements such that $A = R^T R$.*

Proof. The proof is by induction. The result is clearly true for $n = 1$. Assume it is true for $n - 1$. The leading principal submatrix $A_{n-1} = A(1:n-1, 1:n-1)$ is positive definite, so it has a unique Cholesky factorization $A_{n-1} = R_{n-1}^T R_{n-1}$. We have a factorization

$$A = \begin{bmatrix} A_{n-1} & c \\ c^T & \alpha \end{bmatrix} = \begin{bmatrix} R_{n-1}^T & 0 \\ r^T & \beta \end{bmatrix} \begin{bmatrix} R_{n-1} & r \\ 0 & \beta \end{bmatrix} =: R^T R \quad (10.1)$$

if

$$R_{n-1}^T r = c, \quad (10.2)$$

$$r^T r + \beta^2 = \alpha. \quad (10.3)$$

Equation (10.2) has a unique solution since R_{n-1} is nonsingular. Then (10.3) gives $\beta^2 = \alpha - r^T r$. It remains to check that there is a unique real, positive β satisfying this equation. From the equation

$$0 < \det(A) = \det(R^T) \det(R) = \det(R_{n-1})^2 \beta^2$$

we see that $\beta^2 > 0$, hence there is a unique $\beta > 0$. $\square$

The proof of the theorem is constructive, and provides a way to compute the Cholesky factorization that builds R a column at a time. Alternatively, we can work directly from the equations

$$a_{ij} = \sum_{k=1}^i r_{ki} r_{kj}, \quad j \geq i,$$

which follow by equating (i, j) elements in $A = R^T R$. By solving these equations in the order $(1,1)$, $(1,2)$, $(2,2)$, $(1,3)$, $(2,3)$, $(3,3)$, $\dots$, (n,n) , we obtain the following algorithm.

Algorithm 10.2. Given a symmetric positive definite $A \in \mathbb{R}^{n \times n}$ this algorithm computes the Cholesky factorization $A = R^T R$.

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for  $j = 1:n$ 
  for  $i = 1:j-1$ 
     $r_{ij} = (a_{ij} - \sum_{k=1}^{i-1} r_{ki} r_{kj}) / r_{ii}$ 
  end
   $r_{jj} = (a_{jj} - \sum_{k=1}^{j-1} r_{kj}^2)^{1/2}$ 
end

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Cost: $n^3/3$ flops (half the cost of LU factorization).

As for Gaussian elimination (GE), there are different algorithmic forms of Cholesky factorization. Algorithm 10.2 is the jik or “sdot” form. We describe the kij , outer product form in §10.3.

Given the Cholesky factorization $A = R^T R$, a linear system $Ax = b$ can be solved via the two triangular systems $R^T y = b$ and $Rx = y$.

If we define $D = \text{diag}(r_{ii}^2)$ then the Cholesky factorization $A = R^T R$ can be rewritten as $A = LDL^T$, where $L = R^T \text{diag}(r_{ii})^{-1}$ is unit lower triangular. The LDL^T factorization is sometimes preferred over the Cholesky factorization because it avoids the need to compute the n square roots that determine the r_{ii} . The LDL^T factorization is certainly preferred for solving tridiagonal systems, as it requires n less divisions than Cholesky factorization in the substitution stage. All the results for Cholesky factorization in this chapter have analogues for the LDL^T factorization. Block LDL^T factorization for indefinite matrices is discussed in §10.4.

10.1.1. Error Analysis

Error bounds for Cholesky factorization are derived in a similar way to those for LU factorization. Consider Algorithm 10.2. Using Lemma 8.4 we have

$$\left| a_{ij} - \sum_{k=1}^i \hat{r}_{ki} \hat{r}_{kj} \right| \leq \gamma_i \sum_{k=1}^i |\hat{r}_{ki}| |\hat{r}_{kj}|. \quad (10.4)$$

From a variation of Lemma 8.4 in which the division is replaced by a square root (see Problem 10.3), we have

$$\left| a_{jj} - \sum_{k=1}^j \hat{r}_{kj}^2 \right| \leq \gamma_{j+1} \sum_{k=1}^j \hat{r}_{kj}^2.$$

A backward error result is immediate.

Theorem 10.3. *If Cholesky factorization applied to the symmetric positive definite matrix $A \in \mathbf{R}^{n \times n}$ runs to completion then the computed factor R satisfies*

$$\hat{R}^T \hat{R} = A + \Delta A, \quad |\Delta A| \leq \gamma_{n+1} |\hat{R}^T| |\hat{R}|. \quad \square \quad (10.5)$$

Theorem 10.4. *Let $A \in \mathbf{R}^{n \times n}$ be symmetric positive definite and suppose Cholesky factorization produces a computed factor R and a computed solution $\hat{x}$ to $Ax = b$. Then*

$$(A + \Delta A)\hat{x} = b, \quad |\Delta A| \leq 2\gamma_{n+1} |\hat{R}^T| |\hat{R}|. \quad (10.6)$$

Proof. The proof is analogous to the proof of Theorem 9.4. $\square$

These results imply that Cholesky factorization enjoys perfect normwise backward stability. The key inequality is

$$\| |R^T| |R| \|_2 = \| |R| \|_2^2 \leq n \|R\|_2^2 = n \|A\|_2,$$

whose analogue for the computed $\hat{R}$ is, from (10.5),

$$\| |\hat{R}^T| |\hat{R}| \|_2 \leq n(1 - n\gamma_{n+1})^{-1} \|A\|_2.$$

Thus (10.6) implies

$$\|\Delta A\|_2 \leq \| |\Delta A| \|_2 \leq 2\gamma_{n+1} n(1 - n\gamma_{n+1})^{-1} \|A\|_2 \leq 8n(n+1)u \|A\|_2, \quad (10.7)$$

where for the last inequality we have assumed that $n\gamma_{n+1} < 1/2$. Another indicator of stability is that the growth factor for GE is exactly 1 (see Problem 10.4). It is important to realize that the multipliers can be arbitrarily large (consider, for example, $\begin{bmatrix} \theta^2 & \theta \\ \theta & 2 \end{bmatrix}$ as $\theta \rightarrow 0$). But, remarkably, for a positive definite matrix the size of the multipliers has no effect on stability.

Note that the perturbation ΔA in (10.6) is not symmetric, in general, because the backward error matrices for the triangular solves with R and R^T are not the transposes of each other. For conditions guaranteeing that a “small” symmetric ΔA can be found, see Problem 7.11.

The following rewritten version of Theorem 10.3 provides further insight into Cholesky factorization.

Theorem 10.5 (Demmel). *If Cholesky factorization applied to the symmetric positive definite matrix $A \in \mathbb{R}^{n \times n}$ runs to completion then the computed factor $\widehat{R}$ satisfies*

$$\widehat{R}^T \widehat{R} = A + \Delta A, \quad |\Delta A| \leq (1 - \gamma_{n+1})^{-1} \gamma_{n+1} d d^T,$$

where $d_i = a_{ii}^{1/2}$.

Proof. Theorem 10.3 shows that $\widehat{R}^T \widehat{R} = A + \Delta A$ with $|\Delta A| \leq \gamma_{n+1} |\widehat{R}^T| |\widehat{R}|$. Denoting by $\widehat{r}_i$ the i th column of $\widehat{R}$, we have

$$\|\widehat{r}_i\|_2^2 = \widehat{r}_i^T \widehat{r}_i = a_{ii} + \Delta a_{ii} \leq a_{ii} + \gamma_{n+1} |\widehat{r}_i|^T |\widehat{r}_i|,$$

so that $\|\widehat{r}_i\|_2^2 \leq (1 - \gamma_{n+1})^{-1} a_{ii}$. Then, using the Cauchy–Schwarz inequality,

$$|\widehat{r}_i^T \widehat{r}_j| \leq \|\widehat{r}_i\|_2 \|\widehat{r}_j\|_2 \leq (1 - \gamma_{n+1})^{-1} (a_{ii} a_{jj})^{1/2},$$

giving

$$|\widehat{R}^T| |\widehat{R}| \leq (1 - \gamma_{n+1})^{-1} d d^T \quad (10.8)$$

and the required bound for ΔA . $\square$

Standard perturbation theory applied to Theorem 10.4 yields a bound of the form $\|x - \widehat{x}\|/\|x\| \leq c_n u \kappa(A) + O(u^2)$. However, with the aid of Theorem 10.5 we can obtain a potentially much smaller bound. The idea is to write $A = DHD$ where $D = \text{diag}(A)^{1/2}$, so that H has unit diagonal. van der Sluis's result (Corollary 7.6) shows that

$$\kappa_2(H) \leq n \min_{F \text{ diagonal}} \kappa_2(FAF), \quad (10.9)$$

so D is nearly a condition-minimizing diagonal scaling. It follows that $\kappa_2(H) \leq n \kappa_2(A)$ and that $\kappa_2(H) \ll \kappa_2(A)$ is possible if A is badly scaled. Note that $1 \leq \|H\|_2 \leq n$, since H is positive definite with $h_{ii} \equiv 1$.

Theorem 10.6 (Demmel, Wilkinson). *Let $A = DHD \in \mathbb{R}^{n \times n}$ be symmetric positive definite, where $D = \text{diag}(A)^{1/2}$, and suppose Cholesky factorization successfully produces a computed solution $\widehat{x}$ to $Ax = b$. Then the scaled error $D(x - \widehat{x})$ satisfies*

$$\frac{\|D(x - \widehat{x})\|_2}{\|Dx\|_2} \leq \frac{\kappa_2(H)\epsilon}{1 - \kappa_2(H)\epsilon}, \quad (10.10)$$

where $\epsilon := 2n(1 - \gamma_{n+1})^{-1} \gamma_{n+1}$.

Proof. Straightforward analysis shows that (cf. the proof of Theorem 9.4) $(A + \Delta A)\widehat{x} = b$, where

$$\Delta A = \Delta A_1 + \Delta_1 \widehat{R} + \widehat{R}^T \Delta_2 + \Delta_1 \Delta_2,$$

with $|\Delta A_1| \leq (1-\gamma_{n+1})^{-1}\gamma_{n+1}dd^T$ (by Theorem 10.5) and $|\Delta_1| \leq \text{diag}(\gamma_i)|\widehat{R}^T|$, $|\Delta_2| \leq \text{diag}(\gamma_{n-i+1})|\widehat{R}|$. Scaling with D , we have

$$(H + D^{-1}\Delta AD^{-1})D\widehat{x} = D^{-1}b,$$

and standard perturbation theory gives

$$\frac{\|D(x - \widehat{x})\|_2}{\|Dx\|_2} \leq \frac{\kappa_2(H)\|D^{-1}\Delta AD^{-1}\|_2}{1 - \kappa_2(H)\|D^{-1}\Delta AD^{-1}\|_2}.$$

But, using (10.8) and $\|D^{-1}dd^TD^{-1}\|_2 = \|ee^T\|_2 = n$, we have

$$\begin{aligned} \|D^{-1}\Delta AD^{-1}\|_2 &\leq \frac{\gamma_{n+1}}{1 - \gamma_{n+1}}\|D^{-1}dd^TD^{-1}\|_2 \\ &\quad + \text{diag}(\gamma_i + \gamma_{n-i+1} + \gamma_i\gamma_{n-i+1})\|D^{-1}|\widehat{R}^T||\widehat{R}|D^{-1}\|_2 \\ &\leq 2n(1 - \gamma_{n+1})^{-1}\gamma_{n+1}, \end{aligned}$$

using Lemma 3.3, which yields the result. $\square$

Care needs to be exercised when interpreting bounds that involve scaled quantities, but in this case the interpretation is relatively easy. Suppose that H is well conditioned and $\kappa_2(D)$ is large, which represents the artificial ill conditioning that the DHD scaling is designed to clarify. The vector $Dx = H^{-1}D^{-1}b$ is likely to have components that do not vary much in magnitude. Theorem 10.6 then guarantees that we obtain the components of Dx to good relative accuracy and this means that the components of x (which will vary greatly in magnitude) are obtained to good relative accuracy.

So far, our results have all contained the proviso that Cholesky factorization runs to completion—in other words, the results assume that the argument of the square root is always positive. Wilkinson [1092, 1968] showed that success is guaranteed if $20n^{3/2}\kappa_2(A)u \leq 1$, that is, if A is not too ill conditioned. It would be nice to replace A in this condition by H , where $A = DHD$. Justification for doing so is that Algorithm 10.2 is scale invariant, in the sense that if we scale $A \leftarrow FAF$, where F is diagonal, then R scales to RF ; moreover, if F comprises powers of the machine base, then even the rounding errors scale according to F . The following theorem gives an appropriate modification of Wilkinson's condition.

Theorem 10.7 (Demmel). *Let $A = DHD \in \mathbf{R}^{n \times n}$ be symmetric positive definite, when $D = \text{diag}(A)^{1/2}$. If $\lambda_{\min}(H) > n\gamma_{n+1}/(1-\gamma_{n+1})$ then Cholesky factorization applied to A succeeds (barring underflow and overflow) and produces a nonsingular $\widehat{R}$. If $\lambda_{\min}(H) \leq n\gamma_{n+1}/(1-\gamma_{n+1})$ then the computation is certain to fail.*

Proof. The proof is by induction. Consider Algorithm 10.2. The first stage obviously succeeds and gives $\hat{r}_{11} > 0$, since $a_{11} > 0$. Suppose the algorithm has successfully completed $k - 1$ stages, producing a nonsingular R_{k-1} , and consider equations (10.1)-(10.3) with n replaced by k . The k th stage can be completed, but may give a pure imaginary $\hat{\beta}$ (it will if $fl(a - \hat{r}^T \hat{r}) < 0$). However, in the latter event, the error analysis of Theorem 10.5 is still valid! Thus we obtain $\hat{R}_k$ satisfying $\hat{R}_k^* \hat{R}_k = A_k + \Delta A_k$, $|\Delta A_k| \leq (1 - \gamma_{k+1})^{-1} \gamma_{k+1} d_k d_k^T$, where $d_k = [a_{11}^{1/2}, \dots, a_{kk}^{1/2}]^T$. Now, with $D_k = \text{diag}(d_k)$, we have

$$\begin{aligned} \lambda_{\min}(D_k^{-1}(A_k + \Delta A_k)D_k^{-1}) &= \lambda_{\min}(H_k + D_k^{-1}\Delta A_k D_k^{-1}) \\ &\geq \lambda_{\min}(H_k) - \|D_k^{-1}\Delta A_k D_k^{-1}\|_2 \\ &\geq \lambda_{\min}(H_k) - \frac{\gamma_{k+1}}{1 - \gamma_{k+1}} \|ee^T\|_2 \\ &\geq \lambda_{\min}(H) - k \frac{\gamma_{k+1}}{1 - \gamma_{k+1}} > 0, \end{aligned}$$

using the interlacing of the eigenvalues [470, 1989, Cor. 8.1.4] and the condition of the theorem. Hence $D_k^{-1}(A_k + \Delta A_k)D_k^{-1}$ is positive definite, and therefore so is the congruent matrix $A_k + \Delta A_k$, showing that $\hat{R}_k$ must be real and nonsingular, as required to complete the induction.

If Cholesky succeeds, then, by Theorem 10.5, $D^{-1}(A + \Delta A)D^{-1}$ is positive definite and so $0 < \lambda_{\min}(H) + \|D^{-1}\Delta A D^{-1}\|_2 \leq \lambda_{\min}(H) + n(1 - \gamma_{n+1})^{-1} \gamma_{n+1}$. Hence if $\lambda_{\min}(H) < -n\gamma_{n+1}/(1 - \gamma_{n+1})$ then the computation must fail. $\square$

Note that, since $\|H\|_2 \geq 1$, the condition for success of Cholesky factorization can be written as $\kappa_2(H)n\gamma_{n+1}/(1 - \gamma_{n+1}) < 1$.

10.2. Sensitivity of the Cholesky Factorization

The Cholesky factorization has perturbation bounds that are similar to those for LU factorization, but of a simpler form thanks to the positive definiteness ($\|A^{-1}\|_2$ replaces $\|U^{-1}\|_2 \|L^{-1}\|_2$ in the normwise bounds).

Theorem 10.8 (Sun). *Let $A \in \mathbb{R}^{n \times n}$ be symmetric positive definite with the Cholesky factorization $A = R^T R$ and let ΔA be a symmetric matrix satisfying $\|A^{-1} \Delta A\|_2 < 1$. Then $A + \Delta A$ has the Cholesky factorization $A + \Delta A = (R + \Delta R)^T (R + \Delta R)$, where*

$$\frac{\|\Delta R\|_F}{\|R\|_p} \leq 2^{-1/2} \frac{\kappa_2(A)\epsilon}{1 - \kappa_2(A)\epsilon}, \quad \epsilon = \frac{\|\Delta A\|_F}{\|A\|_p}, \quad p = 2, F.$$

Moreover, if $\rho(|\tilde{G}|) < 1$, where $\tilde{G} = (R + \Delta R)^T \Delta A (R + \Delta R)^{-1}$, then

$$|\Delta R| \leq \text{triu}(|\tilde{G}|(I - |\tilde{G}|)^{-1})|R + \Delta R|,$$

where $\text{triu}(\cdot)$ denotes the upper triangular part. $\square$

Note that the Cholesky factor of $A_k = A(1:k, 1:k)$ is R_k , and $\kappa_2(A_{k+1}) \geq \kappa_2(A_k)$ by the interlacing property of the eigenvalues. Hence if A_{k+1} (and hence A) is ill conditioned but A_k is well conditioned then R_k will be relatively insensitive to perturbations in A but the remaining columns of R will be much more sensitive.

10.3. Positive Semidefinite Matrices

If A is symmetric and positive semidefinite ($x^T A x \geq 0$ for all x) then a Cholesky factorization exists, but the theory and computation are more subtle than for positive definite A .

The questions of existence and uniqueness of a Cholesky factorization are answered by the following result.

Theorem 10.9. *Let $A \in \mathbb{R}^{n \times n}$ be positive semidefinite of rank r . (a) There exists at least one upper triangular R with nonnegative diagonal elements such that $A = R^T R$. (b) There is a permutation Π such that $\Pi^T A \Pi$ has a unique Cholesky factorization, which takes the form*

$$\Pi^T A \Pi = R^T R, \quad R = \begin{bmatrix} R_{11} & R_{12} \\ 0 & 0 \end{bmatrix}, \quad (10.11)$$

where R_{11} is $r \times r$ upper triangular with positive diagonal elements.

Proof. (a): Let the symmetric positive semidefinite square root X of A have the QR factorization $X = QR$ with $r_{ii} \geq 0$. Then $A = X^2 = X^T X = R^T Q^T Q R = R^T R$. (b): The algorithm with pivoting described below amounts to a constructive proof. $\square$

Note that the factorization in part (a) is not in general unique. For example,

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \equiv \begin{bmatrix} 0 & 0 \\ \cos \theta & \sin \theta \end{bmatrix} \begin{bmatrix} 0 & \cos \theta \\ 0 & \sin \theta \end{bmatrix}.$$

For practical computations a factorization of the form (10.11) is needed, because this factorization so conveniently displays the rank deficiency. Such a factorization can be computed using an outer product Cholesky algorithm, comprising $r = \text{rank}(A)$ stages. At the k th stage, a rank-1 matrix is subtracted from A so as to introduce zeros into positions $k:n$ in the k th row and

column. Ignoring pivoting for the moment, at the start of the k th stage we have

$$A^{(k)} = (a_{ij}^{(k)}) = A - \sum_{i=1}^{k-1} r_i r_i^T = \begin{matrix} & & k-1 & n-k+1 \\ & & \left[\begin{array}{cc} 0 & 0 \\ 0 & A_k \end{array} \right] \end{matrix}, \quad (10.12)$$

where $r_i^T = [0, \dots, 0, r_{ii}, \dots, r_{in}]$. The reduction is carried one stage further by computing

$$\begin{aligned} r_{kk} &= \sqrt{a_{kk}^{(k)}}, \\ r_{kj} &= a_{kj}^{(k)} / r_{kk}, \quad j = k+1:n, \\ a_{ij}^{(k+1)} &= a_{ij}^{(k)} - r_{ki} r_{kj}, \quad i, j = k+1:n. \end{aligned}$$

Overall we have,

$$A = \sum_{i=1}^r r_i r_i^T = R^T R, \quad R^T = [r_1, \dots, r_r].$$

To avoid breakdown when $a_{kk}^{(k)}$ vanishes (or is negative because of rounding errors), pivoting is incorporated into the algorithm as follows. At the start of the k th stage an element $a_{ss}^{(k)} > 0$ ($s \geq k$) is selected as pivot, and rows and columns k and s of A_k , and the k th and s th elements of r_i , $i = 1:k-1$, are interchanged. The overall effect is to compute the decomposition (10.11), where the permutation Π takes account of all the interchanges.

The standard form of pivoting is defined by

$$s = \min \left\{ j : a_{jj}^{(k)} = \max_{k \leq i \leq n} a_{ii}^{(k)} \right\}.$$

This is equivalent to complete pivoting in GE, since A_k is positive semidefinite so its largest element lies on the diagonal. We note for later reference that this pivoting strategy produces a matrix R that satisfies (cf. Problem 18.5)

$$r_{kk}^2 \geq \sum_{i=k}^{\min(j,r)} r_{ij}^2, \quad j = k+1:n, \quad k = 1:r. \quad (10.13)$$

It will be convenient to denote by $\text{cp}(A) := \Pi^T A \Pi$ the permuted matrix obtained from the Cholesky algorithm with complete pivoting.

10.3.1. Perturbation Theory

In this section we analyse, for a positive semidefinite matrix, the effect on the Cholesky factorization of perturbations in the matrix. This perturbation theory will be used in the error analysis of the next section.

Throughout this section A is assumed to be an $n \times n$ positive semidefinite matrix of rank r whose leading principal submatrix of order r is positive definite. For $k = 1:r$ we will write

$$A = \begin{matrix} & k & n-k \\ \begin{matrix} k \\ n-k \end{matrix} & \begin{bmatrix} A_{11} & A_{12} \\ A_{12}^T & A_{22} \end{bmatrix} \end{matrix} \quad (10.14)$$

and other matrices will be partitioned conformably.

We have the identity

$$A = \begin{matrix} & k \\ \begin{matrix} k \\ n-k \end{matrix} & \begin{bmatrix} R_{11}^T \\ R_{12}^T \end{bmatrix} \end{matrix} [R_{11}, R_{12}] + \begin{bmatrix} 0 & 0 \\ 0 & S_k(A) \end{bmatrix}, \quad (10.15)$$

where R_{11} is the Cholesky factor of A_{11} , $R_{12} = R_{11}^{-T} A_{12}$, and

$$S_k(A) = A_{22} - A_{12}^T A_{11}^{-1} A_{12}$$

is the Schur complement of A_{11} in A . Note that $S_r(A) \equiv 0$, so that for $k = r$, (10.15) is the (unique) Cholesky factorization of A . The following lemma shows how $S_k(A)$ changes when A is perturbed.

Lemma 10.10. *Let E be symmetric and assume $A_{11} + E_{11}$ is nonsingular. Then*

$$S_k(A+E) = S_k(A) + E_{22} - (E_{12}^T W + W^T E_{12}) + W^T E_{11} W + O(\|E\|^2), \quad (10.16)$$

where $W = A_{11}^{-1} A_{12}$.

Proof. We can expand

$$(A_{11} + E_{11})^{-1} = A_{11}^{-1} - A_{11}^{-1} E_{11} A_{11}^{-1} + A_{11}^{-1} E_{11} A_{11}^{-1} E_{11} A_{11}^{-1} + O(\|E_{11}\|^3).$$

The result is obtained by substituting this expansion into $S_k(A+E) = (A_{22} + E_{22}) - (A_{12} + E_{12})^T (A_{11} + E_{11})^{-1} (A_{12} + E_{12})$, and collecting terms. $\square$

Lemma 10.10 shows that the sensitivity of $S_k(A)$ to perturbations in A is governed by the matrix $W = A_{11}^{-1} A_{12}$. The question arises of whether, for a given A , the potential $\|W\|_2^2$ magnification of E indicated by (10.16) is attainable. For the no-pivoting strategy, $\Pi = I$, the answer is trivially “yes”, since we can take $E = \begin{bmatrix} \gamma I & 0 \\ 0 & 0 \end{bmatrix}$, with $|\gamma|$ small, to obtain $\|S_k(A+E) - S_k(A)\|_2 = \|W\|_2^2 \|E\|_2 + O(\|E\|_2^2)$. For complete pivoting, however, the answer is complicated by the possibility that the sequence of pivots will be different for $A+E$ than for A , in which case Lemma 10.10 is not applicable. Fortunately, a mild assumption on A is enough to rule out this technical difficulty, for small $\|E\|_2$. In the next lemma we redefine $A := \text{cp}(A)$ in order to simplify the not at ion.

Lemma 10.11. *Let $A := \text{cp}(A)$. Suppose that*

$$(S_i(A))_{11} > (S_i(A))_{jj}, \quad j = 2:n-i, \quad i = 0:r-1 \quad (10.17)$$

(where $S_0(A) := A$). Then, for sufficiently small $\|E\|_2$, $A+E = \text{cp}(A+E)$. For $E = \begin{bmatrix} \gamma I & 0 \\ 0 & 0 \end{bmatrix}$, with $|\gamma|$ sufficiently small,

$$\|S_k(\text{cp}(A+E)) - S_k(A)\|_2 = \|W\|_2^2 \|E\|_2 + O(\|E\|_2^2).$$

Proof. Note that since $A = \text{cp}(A)$, (10.17) simply states that there are no ties in the pivoting strategy (since $(S_i(A))_{11} \equiv a_{i+1,i+1}^{(i+1)}$ in (10.12)). Lemma 10.10 shows that $S_i(A+E) = S_i(A) + O(\|E\|_2)$, and so, in view of (10.17), for sufficiently small $\|E\|_2$ we have

$$(S_i(A+E))_{11} > (S_i(A+E))_{jj}, \quad j = 2:n-i, \quad i = 0:r-1.$$

This shows that $A + E = \text{cp}(A+E)$. The last part then follows from Lemma 10.10. $\square$

We now examine the quantity $\|W\|_2 = \|A_{11}^{-1}A_{12}\|_2$. We show first that $\|W\|_2$ can be bounded in terms of the square root of the condition number of A_{11} .

Lemma 10.12. *If A , partitioned as in (10.14), is symmetric positive definite and A_{11} is positive definite then $\|A_{11}^{-1}A_{12}\|_2 \leq \sqrt{\|A_{11}^{-1}\|_2 \|A_{22}\|_2}$.*

Proof. Write $A_{11}^{-1}A_{12} = A_{11}^{-1/2}A_{11}^{-1/2}A_{12}$ and use $\|A_{11}^{-1/2}\|_2 = \|A_{11}^{-1}\|_2^{1/2}$, together with $\|A_{11}^{-1/2}A_{12}\|_2 = \|A_{12}^T A_{11}^{-1}A_{12}\|_2^{1/2} \leq \|A_{22}\|_2^{1/2}$, which follows from the fact that the Schur complement $A_{22} - A_{12}^T A_{11}^{-1}A_{12}$ is positive semidefinite. $\square$

Note that, by the arithmetic-geometric mean inequality $\sqrt{xy} \leq (x+y)/2$ ($x, y \geq 0$), we also have, from Lemma 10.12, $\|A_{11}^{-1}A_{12}\|_2 \leq (\|A_{11}^{-1}\|_2 + \|A_{22}\|_2)/2$.

The inequality of Lemma 10.12 is attained for the positive semidefinite matrix

$$A = \begin{bmatrix} \alpha I_{k,k} & I_{k,n-k} \\ I_{n-k,k} & \alpha^{-1} I_{n-k,n-k} \end{bmatrix}, \quad \alpha > 0,$$

where $I_{p,q}$ is the $p \times q$ identity matrix. This example shows that $\|W\|_2$ can be arbitrarily large. However, for $A := \text{cp}(A)$, $\|W\|_2$ can be bounded solely in terms of n and k . The essence of the proof, in the next lemma, is that large elements in A_{11}^{-1} are countered by small elements in A_{12} . Hereafter we set $k = r$, the value of interest in the following sections.

Lemma 10.13. *Let $A := \text{cp}(A)$ and set $k = r$. Then*

$$\|A_{11}^{-1}A_{12}\|_{2,F} \leq \sqrt{\frac{1}{3}(n-r)(4^r-1)}. \quad (10.18)$$

There is a parametrized family of rank- r matrices $A(\theta) = \text{cp}(A(\theta))$, $\theta \in (0, \frac{\pi}{2}]$, for which

$$\|A_{11}(\theta)^{-1}A_{12}(\theta)\|_{2,F} \rightarrow \sqrt{\frac{1}{3}(n-r)(4^r-1)} \quad \text{as } \theta \rightarrow 0.$$

Proof. The proof is a straightforward computation. The matrix $A(O) := R(\theta)^T R(\theta)$, where

$$R(\theta) = \text{diag}(1, s, \dots, s^{r-1}) \begin{bmatrix} 1 & -c & -c & \dots & -c & -c & \dots & -c \\ & 1 & -c & \dots & -c & -c & \dots & -c \\ & & 1 & & \vdots & \vdots & & \vdots \\ & & & \ddots & \vdots & \vdots & & \vdots \\ & & & & 1 & -c & \dots & -c \end{bmatrix} \in \mathbb{R}^{r \times n}, \quad (10.19)$$

with $c = \cos\theta$, $s = \sin\theta$. This is the $r \times n$ version of Kahan's matrix (8.10). R satisfies the inequalities (10.13) (as equalities) and so $A(\theta) = \text{cp}(A(\theta))$. $\square$

We conclude this section with a “worst-case” example for the Cholesky factorization with complete pivoting. Let $U(\theta) = \text{diag}(r, r-1, \dots, 1)R(\theta)$, where $R(\theta)$ is given by (10.19), and define the rank- r matrix $C(\theta) = U(\theta)^T U(\theta)$. Then $C(\theta)$ satisfies the conditions of Lemma 10.11. Also,

$$\begin{aligned} \|W\|_2 &= \|C_{11}(\theta)^{-1}C_{12}(\theta)\|_2 = \|U_{11}(\theta)^{-1}U_{12}(\theta)\|_2 = \|R_{11}(\theta)^{-1}R_{12}(\theta)\|_2 \\ &\rightarrow \sqrt{\frac{1}{3}(n-r)(4^r-1)} \quad \text{as } \theta \rightarrow 0. \end{aligned}$$

Thus, from Lemma 10.11, for $E = \begin{bmatrix} \gamma I & 0 \\ 0 & 0 \end{bmatrix}$, with $|\gamma|$ and θ sufficiently small,

$$\|S_r(\text{cp}(C(\theta) + E))\|_2 \approx \frac{1}{3}(n-r)(4^r-1)\|E\|_2.$$

This example can be interpreted as saying that in exact arithmetic the residual after an r -stage Cholesky factorization of a semidefinite matrix A can overestimate the distance of A from the rank- r semidefinite matrices by a factor as large as $(n-r)(4^r-1)/3$.

10.3.2. Error Analysis

In this section we present a backward error analysis for the Cholesky factorization of a positive semidefinite matrix. An important consideration is that

a matrix of floating point numbers is very unlikely to be “(exactly)” positive semidefinite; errors in forming or storing A will almost certainly render the smallest eigenvalue nonzero, and possibly negative. Therefore error analysis for a rank r positive semidefinite matrix may appear, at first sight, to be of limited applicability. One way around this difficulty is to state results for $A = \tilde{A} + \Delta A$, where $\tilde{A}$ is the matrix stored on the computer, $\tilde{A}$ is positive semidefinite of rank r , and ΔA is a perturbation, which could represent the rounding errors in storing $\tilde{A}$, for example. However, if the perturbation ΔA is no larger than the backward error for the Cholesky factorization, then this extra formalism can be avoided by thinking of ΔA as being included in the backward error matrix. Hence for simplicity, we frame the error analysis for a positive semidefinite A .

The analysis makes no assumptions about the pivoting strategy, so A should be thought of as the pre-permuted matrix $\Pi^T A \Pi$.

Theorem 10.14. *Let A be an $n \times n$ symmetric positive semidefinite matrix of floating point numbers of rank $r \leq n$. Assume that $A_{11} = A(1:r, 1:r)$ is positive definite with*

$$\lambda_{\min}(H_{11}) > r\gamma_{r+1}/(1 - \gamma_{r+1}), \quad (10.20)$$

where $A_{11} = D_{11}H_{11}D_{11}$ and $D_{11} = \text{diag}(A_{11})^{1/2}$. Then, in floating point arithmetic, the Cholesky algorithm applied to A successfully completes r stages (barring underflow and overflow), and the computed $r \times n$ Cholesky factor $\hat{R}_r$ satisfies

$$\|A - \hat{R}_r^T \hat{R}_r\|_2 \leq 2r\gamma_{r+1}\|A\|_2(\|W\|_2 + 1)^2 + O(u^2), \quad (10.21)$$

where $W = A_{11}^{-1}A_{12}$.

Proof. First, note that condition (10.20) guarantees successful completion of the first r stages of the algorithm by Theorem 10.7.

Analysis very similar to that leading to Theorem 10.3 shows that

$$A + E = \hat{R}_r^T \hat{R}_r + \hat{A}^{(r+1)}, \quad (10.22)$$

where

$$\hat{R}_r = \begin{bmatrix} \hat{R}_{11} & \hat{R}_{12} \end{bmatrix}, \quad \hat{A}^{(r+1)} = \begin{bmatrix} 0 & 0 \\ 0 & \hat{S}_{r+1} \end{bmatrix},$$

and

$$|E| \leq \gamma_{r+1}(|\hat{R}_r^T \hat{R}_r| + |\hat{A}^{(r+1)}|). \quad (10.23)$$

Taking norms in (10.23) and using the inequality $\| |B| \|_2 \leq \sqrt{\text{rank}(B)} \|B\|_2$, we obtain

$$\begin{aligned} \|E\|_2 &\leq \gamma_{r+1} (r \|\hat{R}_r^T\|_2 \|\hat{R}_r\|_2 + \sqrt{n-r} \|\hat{A}^{(r+1)}\|_2) \\ &= \gamma_{r+1} (r \|\hat{R}_r^T \hat{R}_r\|_2 + \sqrt{n-r} \|\hat{A}^{(r+1)}\|_2) \\ &= \gamma_{r+1} (r \|A + E - \hat{A}^{(r+1)}\|_2 + \sqrt{n-r} \|\hat{A}^{(r+1)}\|_2) \\ &\leq \gamma_{r+1} (r \|A\|_2 + r \|E\|_2 + n \|\hat{A}^{(r+1)}\|_2), \end{aligned}$$

which implies

$$\|E\|_2 \leq \frac{\gamma_{r+1}}{1 - r\gamma_{r+1}} (r \|A\|_2 + n \|\hat{A}^{(r+1)}\|_2). \quad (10.24)$$

Our aim is to obtain an a priori bound for $\|A - \hat{R}_r^T \hat{R}_r\|_2$. It is clear from (10.22)-(10.24) that to do this we have only to bound $\|\hat{A}^{(r+1)}\|_2$. To this end, we interpret (10.22) in such a way that the perturbation theory of §10.3.1 may be applied.

Equation (10.22) shows that $\hat{S}_{r+1}$ is the true Schur complement for the matrix $A + E$ and that $A_{11} + E_{11} = \hat{R}_{11}^T \hat{R}_{11}$ is positive definite. Hence we can use Lemma 10.10 to deduce that

$$\begin{aligned} \|\hat{A}^{(r+1)}\|_2 &= \|\hat{S}_{r+1}\|_2 \leq \|E_{22}\|_2 + 2\|E_{12}\|_2 \|W\|_2 + \|W\|_2^2 \|E_{11}\|_2 + O(u^2) \\ &\leq \|E\|_2 (\|W\|_2 + 1)^2 + O(u^2). \end{aligned}$$

Substituting from (10.24) we find that

$$\|\hat{A}^{(r+1)}\|_2 \leq r\gamma_{r+1} \|A\|_2 (\|W\|_2 + 1)^2 + O(u^2).$$

Finally, using (10.22) and (10.24), we obtain

$$\|A - \hat{R}_r^T \hat{R}_r\|_2 \leq 2r\gamma_{r+1} \|A\|_2 (\|W\|_2 + 1)^2 + O(u^2). \quad \square$$

Theorem 10.14 is just about the best result that could have been expected, because the bound (10.21) is essentially the same as the bound obtained on taking norms in Lemma 10.10. In other words, (10.21) simply reflects the inherent mathematical sensitivity of $A - R^T R$ to small perturbations in A .

We turn now to the issue of stability. Ideally, for A as defined in Theorem 10.14, the computed Cholesky factor R_r produced after r stages of the algorithm would satisfy

$$\|A - \hat{R}_r^T \hat{R}_r\|_2 \leq c_n u \|A\|_2,$$

where c_n is a modest constant. Theorem 10.14 shows that stability depends on the size of $\|W\|_2 = \|A_{11}^{-1}A_{12}\|_2$ (to the extent that $\|W\|_2$ appears in a realistic bound for the backward error).

If no form of pivoting is used then $\|W\|_2$ can be arbitrarily large for fixed n (see §10.3.1) and the Cholesky algorithm must in this case be classed as unstable. But for complete pivoting we have from Lemma 10.13 the upper bound $\|W\|_2 \leq (1/3(n-r)(4^r-1))^{1/2}$. Thus the Cholesky algorithm with complete pivoting is stable if r is small, but stability cannot be guaranteed, and seems unlikely in practice, if $\|W\|_2$ (and hence, necessarily, r and n) is large.

Numerical experiments show that $\|W\|_2$ is almost always small in practice (typically less than 10) [540, 1990]. However, it is easy to construct examples where $\|W\|_2$ is large. For example, if R is a Cholesky factor of A from complete pivoting then let $C = M(R)^T M(R)$, where $M(R)$ is the comparison matrix; C will usually have a much larger value of $\|W\|_2$ than A .

An important practical issue is when to terminate the Cholesky factorization of a semidefinite matrix. The LINPACK routine `xCHDC` proceeds with the factorization until a nonpositive pivot is encountered, that is, up to and including stage $k-1$, where k is the smallest integer for which

$$\hat{a}_{ii}^{(k)} \leq 0, \quad i = k:n. \quad (10.25)$$

Usually $k > r+1$, due to the effect of rounding errors.

A more sophisticated termination criterion is to stop as soon as

$$\|\hat{S}_k\| \leq \epsilon \|A\| \quad \text{or} \quad \hat{a}_{ii}^{(k)} \leq 0, \quad i = k:n, \quad (10.26)$$

for some readily computed norm and a suitable tolerance ϵ . This criterion terminates as soon as a stable factorization is achieved, avoiding unnecessary work in eliminating negligible elements in the computed Schur complement $\hat{S}_k$. Note that $\|\hat{S}_k\|$ is indeed a reliable order-of-magnitude estimate of the true residual, since $\hat{S}_k$ is the only nonzero block of $\hat{A}^{(k)}$ and, by (10.22) and (10.24), $A - \hat{R}_{k-1}^T \hat{R}_{k-1} = \hat{A}^{(k)} - E$ with $\|E\| = O(u)(\|A\| + \|\hat{A}^{(k)}\|)$.

Another possible stopping criterion is

$$\max_{k \leq i \leq n} \hat{a}_{ii}^{(k)} \leq \epsilon \hat{a}_{11}^{(1)}. \quad (10.27)$$

This is related to (10.26) in that if A (pre-permuted) and $\hat{A}_k$ are positive semidefinite then $\hat{a}_{11}^{(1)} = \max_{i,j} |a_{ij}| \approx \|A\|_2$, and similarly $\max_{k \leq i \leq n} \hat{a}_{ii}^{(k)} \approx \|\hat{S}_k\|_2$. Note that (10.27) bounds $\kappa_2(\hat{R}_{k-1})$, since if (10.27) holds first at the k th stage then, using Theorem 8.13,

$$\frac{\kappa_2(\hat{R}_{k-1})}{(k-1)^{1/2} 2^{k-2}} \leq \frac{|\hat{r}_{11}|}{|\hat{r}_{k-1,k-1}|} = \left(\frac{\hat{a}_{11}^{(1)}}{\hat{a}_{k-1,k-1}^{(k-1)}} \right)^{1/2} \leq \epsilon^{-1/2}.$$

Practical experience shows that the criteria (10.26) and (10.27) with $\epsilon = nu$ both work well, and much more effectively than (10.25) [540, 1990]. We favour (10.27) because of its negligible cost.

10.4. Symmetric Indefinite Matrices and Diagonal Pivoting Method

Let $A \in \mathbb{R}^{n \times n}$ be symmetric but indefinite, that is, $(x^T A x) (y^T A y) < 0$ for some x and y . How can we solve $Ax = b$ efficiently?

Gaussian elimination with partial pivoting (GEPP) can be used to compute the factorization $PA = LU$, but it does not take advantage of the symmetry to reduce the cost and storage. We might try to construct a factorization $A = LDL^T$, where L is unit lower triangular and D is diagonal. But this factorization may not exist, even if symmetric pivoting is allowed, and if it does exist its computation may be unstable. For example, consider

$$A \equiv PAP^T = \begin{bmatrix} \epsilon & 1 \\ 1 & \epsilon \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1/\epsilon & 1 \end{bmatrix} \begin{bmatrix} \epsilon & 0 \\ 0 & \epsilon - 1/\epsilon \end{bmatrix} \begin{bmatrix} 1 & 1/\epsilon \\ 0 & 1 \end{bmatrix}.$$

There is arbitrarily large element growth for $0 < \epsilon \ll 1$, and the factorization does not exist for $\epsilon = 0$.

The most popular approach for solving symmetric indefinite systems is to use a block LDL^T factorization

$$PAP^T = LDL^T,$$

where L is unit lower triangular and D is block diagonal with 1×1 or 2×2 diagonal blocks. This factorization is essentially a symmetric block form of GE, with pivoting. Note that by Sylvester's inertia theorem, A and D have the same inertia¹³, which is easily determined from D (see Problem 10.11).

To begin the computation of the factorization we choose a permutation Π and an integer $s = 1$ or 2 so that

$$\Pi A \Pi^T = \begin{matrix} s \\ n-s \end{matrix} \begin{bmatrix} E & C^T \\ C & B \end{bmatrix},$$

with E nonsingular. Then we compute the factorization

$$\Pi A \Pi^T = \begin{bmatrix} I_s & 0 \\ CE^{-1} & I_{n-s} \end{bmatrix} \begin{bmatrix} E & 0 \\ 0 & B - CE^{-1}C^T \end{bmatrix} \begin{bmatrix} I_s & E^{-1}C^T \\ 0 & I_{n-s} \end{bmatrix}.$$

¹³The inertia of a symmetric matrix is an ordered triple $\{i_+, i_-, i_0\}$, where i_+ is the number of positive eigenvalues, i_- the number of negative eigenvalues, and i_0 the number of zero eigenvalues.

This process is repeated on the $(n - s) \times (n - s)$ Schur complement

$$\tilde{A} = B - CE^{-1}C^T.$$

The cost of the method is $n^3/3$ flops (the same as the cost of Cholesky factorization of a positive definite matrix) plus the cost of determining the permutations 17. This method for computing the block LDL^T factorization is called the *diagonal pivoting method*. It can be thought of as a generalization of Lagrange's method for reducing a quadratic form to diagonal form (devised by Lagrange in 1759 and rediscovered by Gauss in 1823) [763, 1961, p. 371].

One conceivable difficulty with the diagonal pivoting method can be disposed of immediately. If a nonsingular pivot matrix E of dimension 1 or 2 cannot be found, then all 1×1 and 2×2 principal submatrices of the symmetric matrix A are singular, and this is easily seen to imply that A is the zero matrix.

The strategy for choosing 17 is crucial for achieving stability. A suitable modification of the error analysis for block LU factorization (Theorem 12.4) tells us that, provided linear systems involving 2×2 pivots are solved in a normwise backward-stable way, the condition $\|L\| \|D\| \|L^T\| \leq c_n \|A\|$, for a modest constant c_n , is sufficient to ensure stability. A key requirement, therefore, is to choose the pivot E so that the Schur complement $\tilde{A}$ is suitably bounded, since D is made up of elements of Schur complements. We describe two suitable pivoting strategies.

10.4.1. Complete Pivoting

Bunch and Parlett [166, 1971] devised the following strategy for choosing 17. It suffices to describe the interchanges for the first stage of the factorization.

Let $\mu_0 = \max_{i,j} |a_{ij}|$, $\mu_1 = \max_i |a_{ii}|$, and choose $a \in (0, 1)$.
 if $\mu_1 \geq a\mu_0$
 Set $s = 1$, and choose Π so that $|e_{11}| = \mu_1$.
 else
 Set $s = 2$, and choose Π so that $|e_{21}| = \mu_0$.
 end

Note that μ_1 is the best 1×1 pivot under symmetric permutations and μ_0 is the pivot that would be chosen by GE with complete pivoting. This strategy says “as long as there is a diagonal pivot element not much smaller than the complete pivot, choose it as a 1×1 pivot”, that is, “choose a 1×1 pivot whenever possible”. If the strategy dictates the use of a 2×2 pivot then that pivot E is indefinite (see Problem 10.11).

It remains to determine a . This is done by minimizing a bound on the element growth. For the following analysis we assume that the interchanges

have already been done. If $s = 1$ then

$$\tilde{a}_{ij} = b_{ij} - c_{i1} \frac{1}{e_{11}} c_{1j} \Rightarrow |\tilde{a}_{ij}| \leq \mu_0 + \frac{\mu_0^2}{\mu_1} \leq \left(1 + \frac{1}{\alpha}\right) \mu_0.$$

Now consider the case $s = 2$. The (i, j) element of the Schur complement $\tilde{A} = B - CE^{-1}C^T$ is

$$\tilde{a}_{ij} = b_{ij} - [c_{i1} \ c_{i2}] E^{-1} \begin{bmatrix} c_{j1} \\ c_{j2} \end{bmatrix}. \quad (10.28)$$

Now

$$E^{-1} = \begin{bmatrix} e_{11} & e_{12} \\ e_{21} & e_{22} \end{bmatrix}^{-1} = \frac{1}{\det(E)} \begin{bmatrix} e_{22} & -e_{12} \\ -e_{21} & e_{11} \end{bmatrix}$$

and, using the symmetry of E ,

$$\det(E) = e_{11}e_{22} - e_{21}^2 = e_{11}e_{22} - \mu_0^2 \leq \mu_1^2 - \mu_0^2 \leq (\alpha^2 - 1)\mu_0^2.$$

Since $a \in (0, 1)$, we have $|\det(E)| \geq (1 - a^2)\mu_0^2$. Thus

$$|E^{-1}| \leq \frac{1}{(1 - \alpha^2)\mu_0} \begin{bmatrix} \alpha & 1 \\ 1 & \alpha \end{bmatrix}.$$

Since $|c_{ij}| \leq \mu_0$, we obtain from (10.28)

$$|\tilde{a}_{ij}| \leq \mu_0 + \frac{2(1 + \alpha)\mu_0^2}{(1 - \alpha^2)\mu_0} = \left(1 + \frac{2}{1 - \alpha}\right) \mu_0.$$

To determine a we equate the maximum growth for two $s = 1$ steps with that for one $s = 2$ step:

$$\left(1 + \frac{1}{\alpha}\right)^2 = 1 + \frac{2}{1 - \alpha},$$

which reduces to the quadratic equation $4a^2 - a - 1 = 0$. We require the positive root

$$\alpha = \frac{1 + \sqrt{17}}{8} \approx 0.64.$$

The analysis guarantees a growth factor bound of $(1 + a^{-1})^{n-1} = (2.57)^{n-1}$. This bound is pessimistic, however; a much more detailed analysis by Bunch [158, 1971] shows that the growth factor is no more than $3.07(n-1)^{0.446}$ times larger than the bound (9.13) for LU factorization with complete pivoting—a very satisfactory result. Strictly speaking, bounding the growth factor bounds only $\|D\|$, not $\|L\|$. But it is easy to show that for $s = 1$ and 2 no element of CE^{-1} exceeds $\max\{1/a, 1/(1-a)\}$ in absolute value, and so $\|L\|$ is bounded independently of A .

Since complete pivoting requires the whole active submatrix to be searched at each stage, it needs up to $n^3/6$ comparisons, and so the method is rather expensive.

10.4.2. Partial Pivoting

Bunch and Kaufman [164, 1977] devised a pivoting strategy for the diagonal pivoting method that requires only $O(n^2)$ comparisons. At each stage it searches at most two columns and so is analogous to partial pivoting for LU factorization. The strategy contains several logical tests. As before, we describe the pivoting for the first stage only. Recall that s denotes the size of the pivot block.

```

Choose  $a \in (0, 1)$ .
 $\lambda := \|A(2:n, 1)\|_\infty$ 
If  $\lambda = 0$  there is nothing to do on this stage of the elimination.
 $r := \min\{i \geq 2: |a_{i1}| = \lambda\}$ 
if  $|a_{11}| \geq \lambda$ 
(1)  $s = 1, \Pi = I$ 
else
 $\sigma := \left\| \begin{bmatrix} A(1:r-1, r) \\ A(r+1:n, r) \end{bmatrix} \right\|_\infty$ 
if  $|a_{11}|\sigma \geq a\lambda^2$ 
(2)  $s = 1, \Pi = I$ 
else if  $|a_{rr}| > a\sigma$ 
(3)  $s = 1$  and choose  $\Pi$  to swap rows and columns 1 and  $r$ .
else
(4)  $s = 2$  and choose  $\Pi$  to swap rows and columns 2 and  $r$ ,
so that  $|(\Pi A \Pi^T)_{21}| =: \lambda$ .
end
end
end

```

To understand the algorithm it helps to consider the matrix

$$\begin{bmatrix} a_{11} & \dots & \lambda & \dots & \dots & \dots \\ \vdots & & \vdots & & & \\ \lambda & \dots & a_{rr} & \dots & \sigma & \dots \\ \vdots & & \vdots & & & \\ \vdots & & \sigma & & & \\ \vdots & & \vdots & & & \end{bmatrix},$$

and to note that the pivot is one of a_{11} , a_{rr} , and $\begin{bmatrix} a_{11} & \lambda \\ \lambda & a_{rr} \end{bmatrix}$ (or, rather, since $\lambda := |a_{r1}|$, this matrix with λ replaced by a_{r1}).

To bound the element growth reconsider each case in turn, noting that for cases (1) and (2) the elements of the Schur complement are given by¹⁴

$$\tilde{a}_{ij} = a_{ij} - \frac{a_{i1}a_{1j}}{a_{11}}.$$

Case (1):

$$|\tilde{a}_{ij}| \leq |a_{ij}| + \frac{1}{\alpha}|a_{1j}| \leq \left(1 + \frac{1}{\alpha}\right) \max_{i,j} |a_{ij}|.$$

Case (2): Using symmetry,

$$|\tilde{a}_{ij}| \leq |a_{ij}| + \frac{\lambda^2}{|a_{11}|} \leq |a_{ij}| + \frac{\sigma}{\alpha} \leq \left(1 + \frac{1}{\alpha}\right) \max_{i,j} |a_{ij}|.$$

Case (3): The original a_{rr} is now the pivot, and $|a_{rr}| \geq a\sigma$, so

$$|\tilde{a}_{ij}| \leq |a_{ij}| + \frac{\sigma}{|a_{rr}|}|a_{1j}| \leq \left(1 + \frac{1}{\alpha}\right) \max_{i,j} |a_{ij}|.$$

Case (4): This is where we use a 2×2 pivot, which, after the interchanges, is $E = (\Pi^T A \Pi)(1:2, 1:2) = \begin{bmatrix} a_{11} & a_{r1} \\ a_{r1} & a_{rr} \end{bmatrix}$. Now

$$|\det(E)| = |a_{r1}^2 - a_{11}a_{rr}| \geq \lambda^2 - |a_{11}|\alpha\sigma \geq \lambda^2 - \alpha(\alpha\lambda^2) = \lambda^2(1 - \alpha^2).$$

The elements of the Schur complement $\tilde{A} = B - CE^{-1}C^T$ are given by

$$\tilde{a}_{ij} = b_{ij} - \det(E)^{-1} \begin{bmatrix} a_{i1} & a_{ir} \end{bmatrix} \begin{bmatrix} a_{rr} & -a_{r1} \\ -a_{r1} & a_{11} \end{bmatrix} \begin{bmatrix} a_{j1} \\ a_{jr} \end{bmatrix},$$

so

$$\begin{aligned} |\tilde{a}_{ij}| &\leq |b_{ij}| + (\lambda^2(1 - \alpha^2))^{-1} \begin{bmatrix} \lambda & \sigma \end{bmatrix} \begin{bmatrix} \alpha\sigma & \lambda \\ \lambda & |a_{11}| \end{bmatrix} \begin{bmatrix} \lambda \\ \sigma \end{bmatrix} \\ &\leq |b_{ij}| + \frac{2\lambda^2\sigma(1 + \alpha)}{\lambda^2(1 - \alpha^2)} = |b_{ij}| + \frac{2\sigma}{1 - \alpha} \leq \left(1 + \frac{2}{1 - \alpha}\right) \max_{i,j} |a_{ij}|. \end{aligned}$$

This analysis shows that the bounds for the element growth for $s = 1$ and $s = 2$ are the same as the respective bounds for the complete pivoting strategy. Hence, using the same reasoning, we again choose $a = (1 + \sqrt{17})/8$.

The growth factor for the partial pivoting strategy is bounded by $(2.57)^{n-1}$. As for GEPP, large growth factors do not seem to occur in practice. But unlike for GEPP, no example is known for which the bound is attained [164, 1977]; see Problem 10.18.

¹⁴We commit a minor abuse of notation, in that in the rest of this section $\tilde{a}_{ij}$ should really be $\tilde{a}_{i-1, j-1}$ ($s = 1$) or $\tilde{a}_{i-2, j-2}$ ($s = 2$).

As noted in the previous subsection, a bound on the growth factor p_n does not in itself ensure stability. Indeed although $\|D\|/\|A\|$ is bounded for partial pivoting, $\|L\|/\|A\|$ can be arbitrarily large for fixed n ; see Problem 10.15. Higham [559, 1995] gives a detailed error analysis of the diagonal pivoting method with an arbitrary pivoting strategy, under the assumption that computed solutions to linear systems involving 2×2 pivots have a small componentwise relative backward error. The conclusions are that the computed factors satisfy

$$\widehat{L}\widehat{D}\widehat{L}^T = P(A + \Delta A)P^T, \quad \|\Delta A\| \leq p_1(n)u(\|A\| + P^T|\widehat{L}||\widehat{D}||\widehat{L}^T|P) + O(u^2)$$

and the computed solution to a linear system $Ax = b$ satisfies

$$(A + \Delta A)\widehat{x} = b, \quad \|\Delta A\| \leq p_2(n)u(\|A\| + P^T|\widehat{L}||\widehat{D}||\widehat{L}^T|P) + O(u^2),$$

where p_1 and p_2 are linear polynomials. For the partial pivoting strategy, Higham shows that if linear systems involving 2×2 pivots are solved by GEPP or by use of the explicit inverse, then the computed solutions do indeed have a small componentwise relative backward error, and that, moreover,

$$\| |L||D||L^T| \|_M \leq 36np_n\|A\|_M,$$

where $\|A\|_M = \max_{i,j} |a_{ij}|$. Thus the diagonal pivoting method with partial pivoting is stable if the growth factor is small.

10.5. Nonsymmetric Positive Definite Matrices

The notion of positive definiteness can be extended to nonsymmetric matrices. A nonsymmetric matrix $A \in \mathbb{R}^{n \times n}$ is *positive definite* if $x^T Ax > 0$ for all $x \neq 0$. This is equivalent to the condition that the symmetric part A_S of A is positive definite, where $A \equiv A_S + A_K$ with $A_S = (A + A^T)/2$ and $A_K = (A - A^T)/2$. A positive definite matrix clearly has nonsingular leading principal submatrices, and so has an LU factorization, $A = LU$. It can even be shown that pivots u_{ii} are positive. However, there is no guarantee that the factorization is stable without pivoting, as the example $\begin{bmatrix} \epsilon & 1 \\ -1 & \epsilon \end{bmatrix}$ shows. The standard error analysis for LU factorization applies (Theorems 9.3 and 9.4), and so the question is whether $|\widehat{L}||\widehat{U}|$ can be suitably bounded. Golub and Van Loan [469, 1979] show that, for the exact LU factors,

$$\| |L||U| \|_F \leq n\|A_S + A_K^T A_S^{-1} A_K\|_2. \quad (10.29)$$

Let $\chi(A) = \|A_S + A_K^T A_S^{-1} A_K\|_2 \|A_S^{-1}\|_2$, which is just $\kappa_2(A)$ when A is symmetric. Mathias [731, 1992] shows that $\| |L||U| \|_F$ (involving now the computed LU factors) is at most a factor $1 + 30un^{3/2}\chi(A)$ times larger than the

upper bound in (10.29), and that the LU factorization (without pivoting) succeeds if $24n^{3/2}X(A)u \leq 1$.

These results show that it is safe not to pivot provided that the symmetric part of A is not too ill conditioned relative to the norm of the skew-symmetric part. If A is symmetric ($A_K = 0$) then we recover the results for symmetric positive definite matrices.

10.6. Notes and References

André-Louis Cholesky (1875-1918) was a French military officer involved in geodesy and surveying in Crete and North Africa. In some books his name is misspelled “Choleski”. Details of Cholesky’s life and a discussion about the pronunciation of his name!-can be found in the electronic mail magazine NA-Digest, volume 90, 1990, issues 7, 8, 10–12, and 24; see, in particular, the biography [22, 1922]. Cholesky’s work was published posthumously on his behalf by Benoit [91, 1924].

The properties of the Cholesky factorization are intimately associated with the properties of the Schur complement, as is apparent from some of the proofs in this chapter. The same is true for GE in general. An excellent survey of the Schur complement, containing historical comments, theory, and applications, is given by Cottle [248, 1974].

For results on the Cholesky factorization in Hilbert space see Power [841, 1986].

A book by George and Liu [438, 1981] is devoted to the many practical issues in the implementation of Cholesky factorization for the solution of sparse symmetric positive definite systems.

There is no floating point error analysis of Cholesky factorization in Wilkinson’s books, but he gives a detailed analysis in [1092, 1968], showing that $\widehat{R}^T\widehat{R} = A + E$, with $\|E\|_2 \leq 2.5n^{3/2}u\|A\|_2$. It is unfortunate that this paper is in a rather inaccessible proceedings, because it is a model of how to phrase and interpret an error analysis. Meinguet [747, 1983] and Sun [973, 1992] give componentwise backward error bounds similar to those in Theorems 10.3 and 10.4. Kielbasinski [657, 1987] reworks Wilkinson’s analysis to improve the constant.

The fact that $\kappa_2(H)$ can replace the potentially much larger $\kappa_2(A)$ in the forward error bound for the Cholesky method was stated informally and without proof by Wilkinson [1092, 1968, p. 638]. Demmel [283, 1989] made this observation precise and explored its implications; Theorems 10.5, 10.6, and 10.7 are taken from [283, 1989].

The bounds in Theorem 10.8 are from Sun [971, 1991], [972, 1992]. Similar bounds are given by Stewart [944, 1977], [951, 1993], Barrlund [71, 1991], and Sun [973, 1992]. A perturbation bound that can be much smaller than

the normwise one in Theorem 10.8 is derived and explored by Chang and Paige [198, 1995]. Perturbation results of a different flavour, including one for structured perturbations of the form of ΔA in Theorem 10.5, are given by Drmac, Omladic, and Veselic [321, 1994].

The perturbation and error analysis of §10.3 for semidefinite matrices is from Higham [540, 1990], where in a perturbation result for the QR factorization with column pivoting is also given. For an application in optimization that makes use of Cholesky factorization with complete pivoting and the analysis of §10.3.1 see Forsgren, Gill, and Murray [384, 1995].

Fletcher and Powell [382, 1974] describe several algorithms for updating an LDL^T factorization of a symmetric positive definite A when A is modified by a rank-1 matrix. They give detailed componentwise error analysis for some of the methods.

An excellent way to test whether a given symmetric matrix A is positive (semi) definite is to attempt to compute a Cholesky factorization. This test is less expensive than computing the eigenvalues and is numerically stable. Indeed, if the answer “yes” is obtained, it is the right answer for a nearby matrix, whereas if the answer is “no” then A *must* be close to an indefinite matrix. See Higham [535, 1988] for an application of this definiteness test. An algorithm for testing the definiteness of a Toeplitz matrix is developed by Cybenko and Van Loan [260, 1986], as part of a more complicated algorithm. According to Kerr [654, 1990], misconceptions of what is a sufficient condition for a matrix to be positive (semi) definite are rife in the engineering literature (for example, that it suffices to check the definiteness of all 2×2 submatrices). See also Problem 10.8. For some results on definiteness tests for Toeplitz matrices, see Makhoul [722, 1991].

A major source of symmetric indefinite linear systems is the least squares problem, because the augmented system is symmetric indefinite; see Chapter 19. Other sources of such systems are interior methods for solving constrained optimization problems (see Forsgren, Gill, and Shinnerl [385, 1996], Turner [1030, 1991], and Wright [1115, 1992]) and linearly constrained optimization problems (see Gill, Murray, Saunders, and Wright [445, 1990], [446, 1991]).

The idea of using a block LDL^T factorization with some form of pivoting for symmetric indefinite matrices was first suggested by Kahan in 1965 [166, 1971]. Bunch and Parlett [166, 1971] developed the complete pivoting strategy and Bunch [158, 1971] proved its stability. Bunch [160, 1974] discusses a rather expensive partial pivoting strategy that requires repeated scalings. Bunch and Kaufman [164, 1977] found the efficient partial pivoting strategy presented here, which is the one now widely used, and Bunch, Kaufman and Parlett [165, 1976] give an Algol code implementing the diagonal pivoting method with this pivoting strategy. Dongarra, Duff, Sorensen, and van der Vorst [315, 1991, §5.4.5] show how to develop a partitioned version of the diagonal pivoting

method with partial pivoting.

Liu [709, 1987] shows how to incorporate a threshold into the Bunch–Kaufman partial pivoting strategy for sparse symmetric matrices; see also Duff et al. [326, 1991]. The partial pivoting strategy and variants of it described by Bunch and Kaufman [164, 1977] do not preserve band structure, but the fill-in depends on the number of 2×2 pivots, which is bounded by the number of negative eigenvalues (see Problem 10.11). Jones and Patrick [615, 1993], [616, 1994] show how to exploit this fact.

The complete and partial pivoting strategies of Bunch et al. use a fixed number of tests to determine each pivot. Another possibility is to prescribe growth bounds corresponding to 1×1 and 2×2 pivots and to search in some particular order for a pivot satisfying the bound. Fletcher [375, 1976] uses this approach to define a pivoting strategy that usually requires only $O(n^2)$ operations. Duff, Reid, and co-workers apply the same approach to the diagonal pivoting method for sparse matrices, where sparsity considerations also influence the choice of pivot [331, 1979], [326, 1991]; their Fortran codes MA27 [329, 1982] and MA47 [330, 1995] implement the methods.

Gill, Murray, Ponceleón, and Saunders [443, 1992] show how for sparse, symmetric indefinite systems the diagonal pivoting factorization can be used to construct a (positive definite) preconditioned for an iterative method.

Another method for solving symmetric indefinite systems is Aasen’s method [1, 1971], which employs the factorization $PAP^T = LTL^T$, where L is unit lower triangular and T is tridiagonal. It is competitive with the diagonal pivoting method in terms of speed. Barwell and George [77, 1976] compare the performance of Fortran codes for several methods for solving symmetric indefinite systems, including the diagonal pivoting method and Aasen’s method.

Dax and Kaniel [270, 1977] propose computing a factorization $PAP^T = LDL^T$ for symmetric indefinite matrices by an extended form of Gaussian elimination in which extra row operations are used to “build up” a pivot element prior to the elimination operations; here, L is unit lower triangular and D is diagonal. A complete pivoting strategy for determining the permutation P is described in [270, 1977] and partial pivoting strategies in Dax [268, 1982].

Analogues of the factorization for symmetric matrices exist for skew-symmetric matrices; see Bunch [161, 1982].

Bunch [159, 1971] shows how to scale a symmetric matrix so that in every nonzero row and column the largest magnitude of an element is 1.

10.6.1. LAPACK

Driver routines `xPOSV` (simple) and `xPOSVX` (expert) use the Cholesky factorization to solve a symmetric (or Hermitian) positive definite system of linear equations with multiple right-hand sides. (There are corresponding

routines for packed storage, in which one triangle of the matrix is stored in a one-dimensional array: PP replaces PO in the names.) The expert driver incorporates iterative refinement, condition estimation, and backward and forward error estimation and has an option to scale the system $AX = B$ to $(D^{-1}AD^{-1})DX = D^{-1}B$, where $D = \text{diag}(a_{ii}^{1/2})$. Modulo the rounding errors in computing and applying the scaling, the scaling has no effect on the accuracy of the solution prior to iterative refinement, in view of Theorem 10.6. The Cholesky factorization is computed by the routine `xPOTRF`, which uses a partitioned algorithm that computes R a block row at a time. The drivers `xPTSV` and `xPTSVX` for symmetric positive definite tridiagonal matrices use LDL^T factorization. LAPACK does not currently contain a routine for Cholesky factorization of a positive semidefinite matrix, but there is such a routine in LINPACK (`xCHDC`).

Driver routines `xsYSV` (simple) and `xsYSVX` (expert) use the block LDL^T factorization (computed by the diagonal pivoting method) with partial pivoting to solve a symmetric indefinite system of linear equations with multiple right-hand sides. For Hermitian matrices the corresponding routines are `xHESV` (simple) and `xHESVX` (expert). (Variants of these routines for packed storage have names in which SP replaces SY and HP replaces HE.) The expert drivers incorporate iterative refinement, condition estimation, and backward and forward error estimation. The factorization is computed by the routine `xsYTRF` or `xHETRF`.

Problems

10.1. Show that if $A \in \mathbb{R}^{n \times n}$ is symmetric positive definite then

$$|a_{ij}| < \sqrt{a_{ii}a_{jj}} \quad \text{for all } i \neq j.$$

What does this statement imply about $\max_{i,j} |a_{ij}|$?

10.2. If A is a symmetric positive definite matrix, how would you compute $x^T A^{-1} x$?

10.3. Let $y = (c - \sum_{i=1}^{k-1} a_i b_i)^{1/2}$ be evaluated in floating point arithmetic in any order. Show that

$$\hat{y}^2(1 + \theta_{k+1}) = c - \sum_{i=1}^{k-1} a_i b_i (1 + \theta_{k-1}^{(i)}),$$

where $|\theta_{k-1}^{(i)}| \leq \gamma_{k-1}$ for all i , and $|\theta_{k+1}| \leq \gamma_{k+1}$.

10.4. Let $A \in \mathbb{R}^{n \times n}$ be symmetric positive definite. Show that the reduced submatrix B of order $n-1$ at the end of the first stage of GE is also symmetric